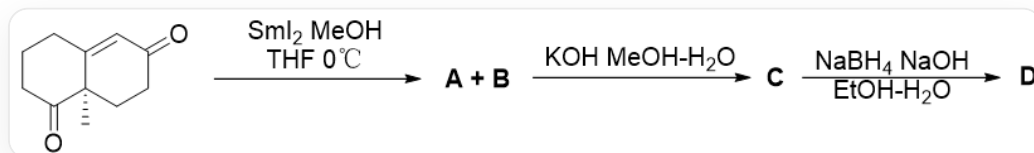


Question

Observe the following reaction:

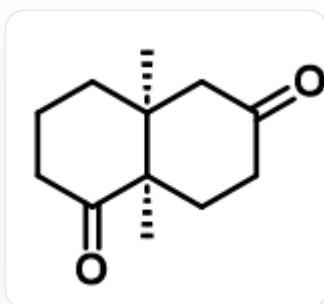


O=C([C@]1(C)CC2)CCCC1=CC2=O and Sml_2 react in THF/MeOH solvent at 0°C to obtain **A** and **B**, both of which can be converted to **C** under KOH , $\text{MeOH}/\text{H}_2\text{O}$ conditions. **C** can be converted to **D** under NaBH_4 , NaOH , $\text{EtOH}/\text{H}_2\text{O}$ conditions.

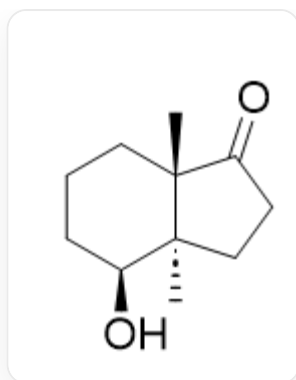
It is known that both **A** and **B** can be converted to compound **C**, and that both **A** and **B** as reaction substrates can yield **C** with similar yields. Both products **A** and **B** contain a three-membered ring. In the ^1H NMR spectrum of **C**, there are two peaks whose integral area corresponds to 3H .

Which of the following statements is correct:

- A** and **B** cannot be converted to each other in the presence of KOH
- The structure of **C** is shown in the figure



- Given that **D** has one ketone carbonyl group, then the structure of **D** should be as shown in the figure



C[C@@]12[C@](CCC2=O)(C)[C@@H](O)CCC1

4. The reason why only one carbonyl group can be selectively reduced from **C** to **D** is that there is a difference in steric hindrance around the two carbonyl groups.

A. All other options are incorrect

B. 2.4.

C. 1.3.

D. 1.2.4.

E. 3.4.

F. 2.3.4.

G. 3.

Answer

Correct Answer: **G**

Detailed Explanation

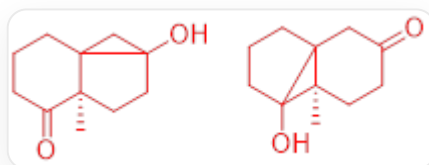
It is known that both **A** and **B** can be converted into compound **C**, and whether **A** or **B** is used as the reaction substrate, **C** can be obtained with similar yields, indicating that **A** and **B** can be interconverted in the presence of KOH. 1 Incorrect

CHECKPOINT

1 PTS

A and **B** can be interconverted in the presence of KOH

Therefore, its structure can be deduced:



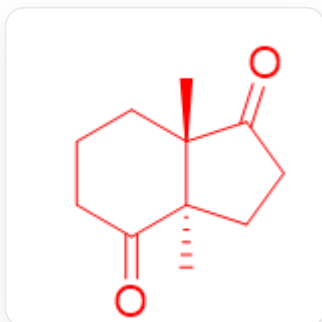
O=C1CCCC23[C@@]1(C)CCC(O)2C3 OC12CCCC31[C@@]2(C)CCC(C3)=O

CHECKPOINT

2 PTS

The structure of **A** is O=C1CCCC23[C@@]1(C)CCC(O)2C3, and the structure of **B** is OC12CCCC31[C@@]2(C)CCC(C3)=O

The former can undergo ring-opening to obtain the hexafused-pentacyclic structure **C**. Contains two methyl groups. 2 Incorrect



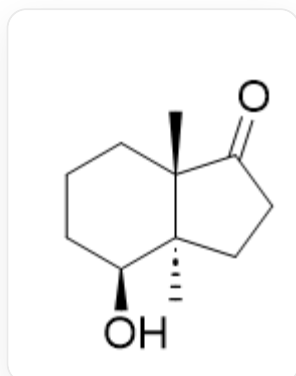
C[C@@]12[C@](CCC2=O)(C)C(CCC1)=O

CHECKPOINT

1 PTS

The structure of **C** is C[C@@]12[C@](CCC2=O)(C)C(CCC1)=O

The hexacyclic carbonyl group is reduced more quickly because it experiences less torsional strain during the reduction process, yielding **D**



C[C@@]12[C@](CCC2=O)(C)[C@@H](O)CCC1

CHECKPOINT

1 PTS

The hexacyclic carbonyl group is reduced more quickly because it experiences less torsional strain during the reduction process

CHECKPOINT

1 PTS

The structure of **D** is C[C@@]12[C@](CCC2=O)(C)[C@@H](O)CCC1

3 Correct, 4 Incorrect.

G is correct.